

Sprint 7 – Bulk Processing and Governor Limits Report

Experiment 1 – Governor Limits

Objective

To understand Salesforce Governor Limits and learn how SOQL queries are limited per transaction.

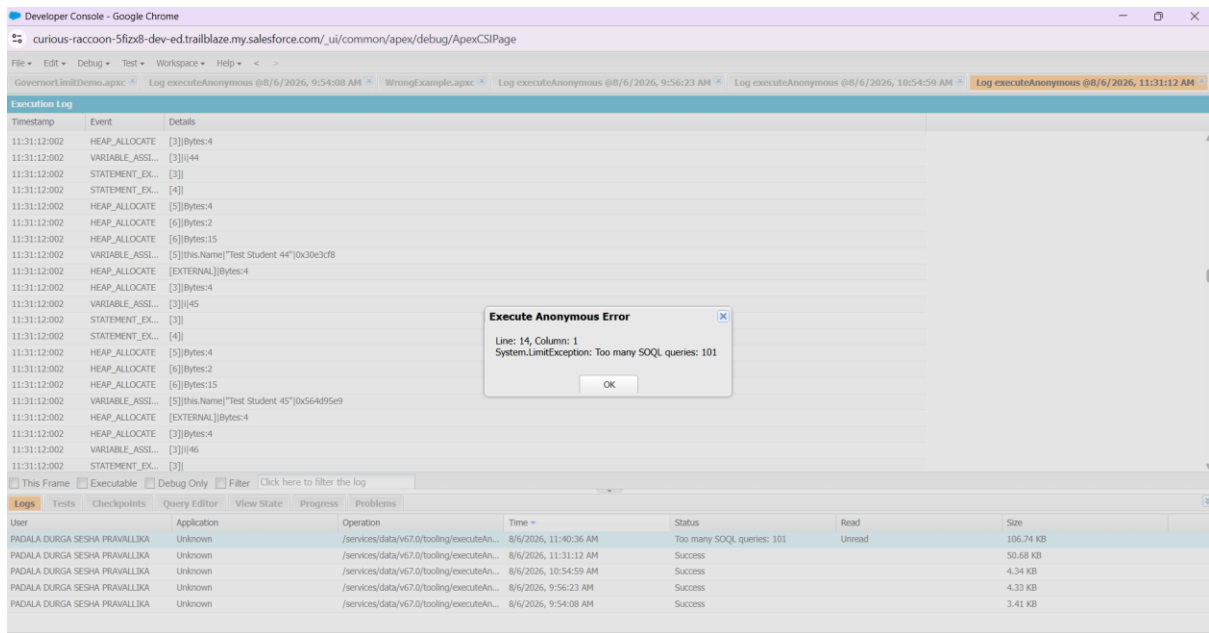
Code

```
public class WrongExample {  
    public static void testWrong() {  
        List<Student__c> students = [  
            SELECT Id, Name  
            FROM Student__c  
        ];  
        for(Student__c st : students){  
            Student__c s = [  
                SELECT Id, Name  
                FROM Student__c  
                WHERE Id = :st.Id  
            ];  
            System.debug(s.Name);  
        }  
    }  
}
```

Execute

```
WrongExample.testWrong();
```

Output



Experiment 2 – Bulkified SOQL

Objective

To understand how to write bulkified SOQL queries outside loops to efficiently process multiple records while avoiding Salesforce Governor Limit exceptions.

Code

```
public class CollectionExample {

    public static void testCollection() {

        List<Student__c> students = [

            SELECT Id, Name

            FROM Student__c

        ];

        for(Student__c s : students){

            System.debug(s.Name);

        }

    }

}
```

Execute

```
CollectionExample.testCollection();
```

Output

Developer Console - Google Chrome

curious-raccoon-5fizx8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >

CollectionExample.apxc Log executeAnonymous @8/6/2026, 11:52:05 AM

Execution Log

Timestamp	Event	Details
11:52:05:026	CUMULATIVE_L...	
11:52:05:026	LIMIT_USAGE_...	(default)
11:52:05:000	LIMIT_USAGE_...	Number of SOQL queries: 1 out of 100
11:52:05:000	LIMIT_USAGE_...	Number of query rows: 106 out of 50000
11:52:05:000	LIMIT_USAGE_...	Number of SOSL queries: 0 out of 20
11:52:05:000	LIMIT_USAGE_...	Number of DML statements: 0 out of 150
11:52:05:000	LIMIT_USAGE_...	Number of Publish Immediate DML: 0 out of 150
11:52:05:000	LIMIT_USAGE_...	Number of DML rows: 0 out of 10000
11:52:05:000	LIMIT_USAGE_...	Maximum CPU time: 0 out of 10000
11:52:05:000	LIMIT_USAGE_...	Maximum heap size: 0 out of 6000000
11:52:05:000	LIMIT_USAGE_...	Number of callouts: 0 out of 100
11:52:05:000	LIMIT_USAGE_...	Number of Email Invocations: 0 out of 10
11:52:05:000	LIMIT_USAGE_...	Number of future calls: 0 out of 50
11:52:05:000	LIMIT_USAGE_...	Number of queueable jobs added to the queue: 0 out of 50
11:52:05:000	LIMIT_USAGE_...	Number of Mobile Apex push calls: 0 out of 10
11:52:05:000	LIMIT_USAGE_...	
11:52:05:026	CUMULATIVE_L...	

☐ This Frame ☐ Executable ☐ Debug Only ☒ Filter LIMIT_USAGE

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA ...	Unknown	/services/data/v6...	8/6/2026, 11:52:...	Success		48.14 KB

☐ Filter Click here to filter the log list

Experiment 3 – List Example

Objective

To learn how to create, store, access, and iterate records using the List collection in Apex.

Code

```
public class ListExample {
    public static void testList() {
        List<Student__c> students = [
            SELECT Id, Name
```

```

FROM Student__c

];

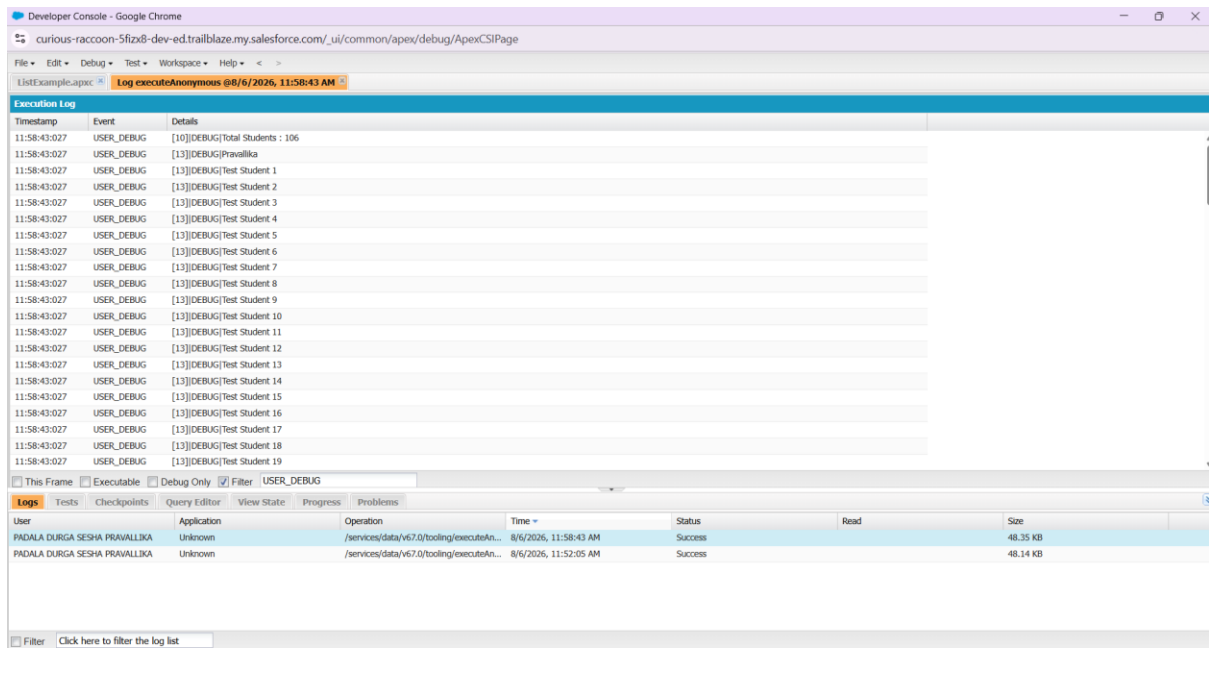
System.debug(students.size());

}

}

```

Output



The screenshot shows the Salesforce Developer Console interface. The top part displays the 'Execution Log' tab, which lists a series of 'USER_DEBUG' events. Each event contains a message indicating the total number of students (106) and a list of student IDs (1 through 19). The bottom part shows the 'Logs' tab, which displays a table of log entries. The table has columns for User, Application, Operation, Time, Status, Read, and Size. Two log entries are visible, both for the user 'PADALA DURGA SETHA PRAVALLIKA'. The first entry is for the operation '/services/data/v67.0/tooling/executeAnonymous' and the second is for '/services/data/v67.0/tooling/executeAnonymous'. Both entries are marked as 'Success'.

Timestamp	Event	Details
11:58:43:027	USER_DEBUG	[10]DEBUG Total Students : 106
11:58:43:027	USER_DEBUG	[13]DEBUG Pravallika
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 1
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 2
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 3
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 4
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 5
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 6
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 7
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 8
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 9
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 10
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 11
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 12
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 13
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 14
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 15
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 16
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 17
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 18
11:58:43:027	USER_DEBUG	[13]DEBUG Test Student 19

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SETHA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeAnonymous	8/6/2026, 11:58:43 AM	Success		48.35 KB
PADALA DURGA SETHA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeAnonymous	8/6/2026, 11:52:05 AM	Success		48.14 KB

Experiment 4 – Set Example

Objective

To understand how Sets remove duplicate values and improve data processing performance.

Code

```

public class SetExample {

    public static void testSet() {

        Set<Id> ids = new Set<Id>();

        for(Student__c s :

            [SELECT Id FROM Student__c]){

                ids.add(s.Id);

            }

        System.debug(ids.size());

    }

}

```

}

Output

The screenshot shows the Salesforce Developer Console interface. The top bar indicates the user is logged in as 'curious-raccoon-5f1zx8-dev-ed.trailblaze.my.salesforce.com'. The main area displays the 'Execution Log' for the file 'SetExample.apex'. The log shows a single event: 'USER_DEBUG' at '12:00:42:030' with the message '[16]DEBUG:Unique Student IDs : 106'. Below the log, the 'Logs' tab is selected, showing a table of log entries.

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SETHA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeAn...	8/6/2026, 12:00:42 PM	Success		45.38 KB
PADALA DURGA SETHA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeAn...	8/6/2026, 11:58:43 AM	Success		48.35 KB
PADALA DURGA SETHA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeAn...	8/6/2026, 11:52:05 AM	Success		48.14 KB

Experiment 5 – Map Example

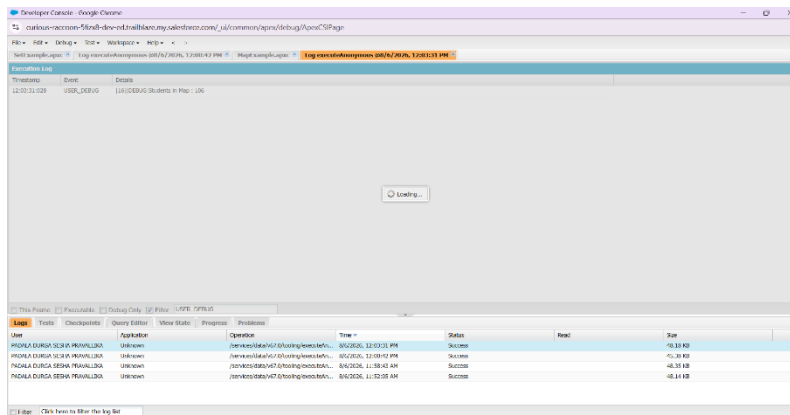
Objective

To learn how Maps store key-value pairs and enable fast record retrieval in Apex.

Code

```
public class MapExample {  
    public static void testMap() {  
        Map<Id,Student__c> studentMap =  
            new Map<Id,Student__c>(  
                [SELECT Id,Name FROM Student__c]  
            );  
        System.debug(studentMap.size());  
    }  
}
```

Output



Experiment 6 – Bulk Processing Pattern

Objective

To learn how to process multiple records efficiently using bulkified Apex code and Trigger.new collections.

Code

```
public class BulkProcessingExample {

    public static void testBulkProcessing() {

        List<Student__c> students = [

            SELECT Id, Name

            FROM Student__c

        ];

        Set<Id> studentIds = new Set<Id>();

        for(Student__c s : students){

            studentIds.add(s.Id);

        }

        Map<Id, Student__c> studentMap =

            new Map<Id, Student__c>(

                [SELECT Id, Name

                FROM Student__c

                WHERE Id IN :studentIds]

            );

        for(Id sid : studentMap.keySet()){

            Student__c stu = studentMap.get(sid);
```

```

        System.debug('Student Name : ' + stu.Name);
    }

    List<Student__c> studentsToUpdate =

        new List<Student__c>();

    for(Student__c s : students){

        studentsToUpdate.add(s);

    }

    System.debug('Records Ready for Update : '

        + studentsToUpdate.size());

}

}

```

Output

The screenshot shows the Salesforce Developer Console. The top part displays the 'Execution Log' with 105 entries, each showing a timestamp of 12:07:45:054, the event 'USER_DEBUG', and the details '[29]DEBUG|Student Name : Test Student' followed by a number from 81 to 105. The bottom part shows the 'Logs' tab with three entries, all of which are successful executions of the 'executeAnonymous' operation, each with a size of 175.14 KB.

Timestamp	Event	Details
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 81
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 82
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 83
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 84
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 85
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 86
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 87
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 88
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 89
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 90
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 91
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 92
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 93
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 94
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 95
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 96
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 97
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 98
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 99
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 100
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 101
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 102
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 103
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 104
12:07:45:054	USER_DEBUG	[29]DEBUG Student Name : Test Student 105

User	Application	Operation	Time	Status	Read	Size
PADALA DURGA SESHU PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeA...	8/6/2026, 12:07:45 PM	Success		175.14 KB
PADALA DURGA SESHU PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeA...	8/6/2026, 12:03:31 PM	Success		48.18 KB
PADALA DURGA SESHU PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeA...	8/6/2026, 12:00:43 PM	Success		45.38 KB

Experiment 7 – DMLLoopExample

Objective

To understand why DML statements should not be placed inside loops and learn the bulk DML approach.

Code

```

public class DMLLoopExample {

    public static void testWrongDML() {

```

```

List<Student__c> students = [

    SELECT Id, Name

    FROM Student__c

];

for(Student__c s : students){

    s.Name = s.Name + ' Test';

    update s;

}

}

```

Output

The screenshot shows the Salesforce Developer Console interface. The top part displays the Execution Log with various system events and their details. The bottom part shows the Logs tab with a table of log entries.

User	Application	Operation	Time	Status	Read	Size
PADALA DURIGA SESHIA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeA...	8/6/2026, 12:11:50 PM	Success		84.86 KB
PADALA DURIGA SESHIA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeA...	8/6/2026, 12:07:45 PM	Success		175.14 KB
PADALA DURIGA SESHIA PRAVALLIKA	Unknown	/services/data/v67.0/tooling/executeA...	8/6/2026, 12:03:31 PM	Success		40.18 KB

Experiment 8 – DMLBulkExample

Objective

To understand how to perform bulk DML operations in Apex by updating multiple records in a single DML statement, thereby improving performance and avoiding Governor Limit exceptions.

Code

```

public class DMLBulkExample {

    public static void testCorrectDML() {

        List<Student__c> students = [

```



```

SELECT Id, Name

FROM Student__c

];

for(Student__c s : students){

    s.Name = s.Name + ' Updated';

}

update students;

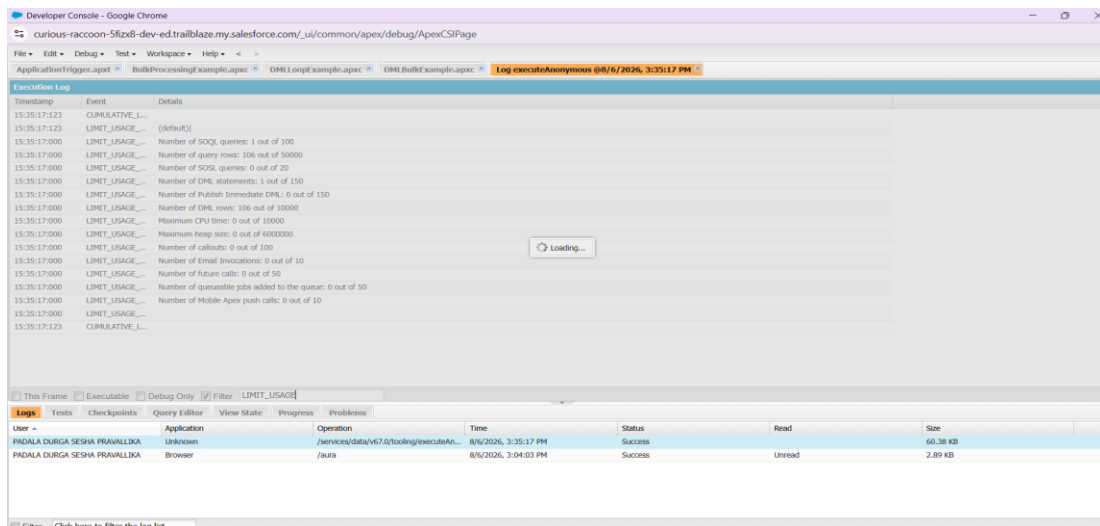
System.debug('Students Updated : ' + students.size());

}

}

```

Output



Experiment 9 – Receive the Applications

Objective

To create a bulkified Apex Trigger that processes multiple Application records efficiently while following Salesforce governor limits and best practices.

Trigger Code

```

trigger ApplicationTrigger on Application__c (before insert) {

    List<Application__c> applications = Trigger.new;

    System.debug('Total Applications = ' + applications.size());

}

```

Output

Developer Console - Google Chrome

curious-raccoon-5f2a8-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSiPage

File • Edit • Debug • Test • Workspace • Help • < • >

ApplicationTrigger.apex • BulkProcessingSample.apex • DMHLogSample.apex • DMHBuildSample.apex • Log executeAnonymous (6/6/2026, 3:35:17 PM) • **Log executeAnonymous (6/6/2026, 3:04:03 PM)**

Execution Log

Timestamp	Event	Details
15:04:03:006	USER_DEBUG	[7]DEBUG:Total Applications = 1

⌂ Loading...

This Frame • Executable • Debug Only • Filter: USER_DEBUG

Logs • Tools • Checkpoints • Query Editor • View State • Progress • Problems

User	Application	Operation	Time	Status	Read	Size
PAHALA DURGA SESHU PRASADLAKA	Unknown	/services/data/v47.0/tracking/executeAn...	6/6/2026, 3:35:17 PM	Success		60.38 KB
PAHALA DURGA SESHU PRASADLAKA	Browser	/aura	6/6/2026, 3:04:03 PM	Success		2.89 KB

Filter: Click here to filter the log list